McMaster University ENGINEERING | mohawk | W BOOTH SCHOOL OF ENGINEERING PRACTICE AND TECHNOLOGY
McMaster-Mohawk Bachelor of Technology Partnership

Laboratory 7
Introduction to Ubidots

SMRTTECH 3CC3

Cloud Computing and Internet of Things

B. Tech - SMRTTECH

OBJECTIVES

In this lab you will learn the following:

- Use an **Ultrasonic Sensor** to determine the distance of an object from the device.
- Use **MQTT protocol** to transfer data between a LabVIEW Client and Ubidots IoT platform.
- Create a web hosted dashboard that can be used for monitoring and control.

The main outcome of this lab is to send sensor data to the cloud and to control a simple actuator using the cloud.

MATERIALS

- Arduino Mega
- Ultrasonic Sensor
- LED
- 220 Ohm Resistor
- Jumper Wires

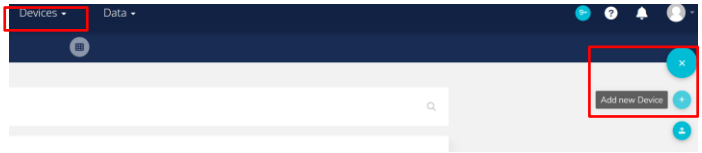
PROCEDURE

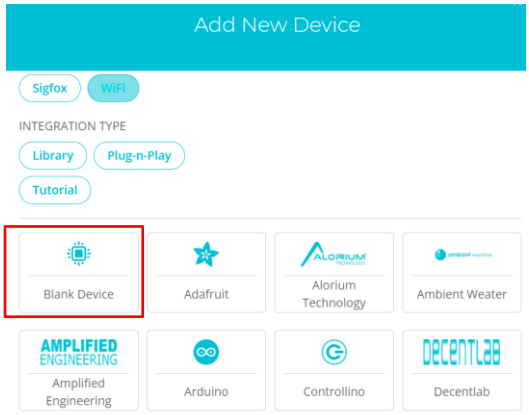
Exercise 1: Creating a Ubidots Account and an IoT device

1. Go to the following to the following website and sign-up for a Ubidots stem account for free.

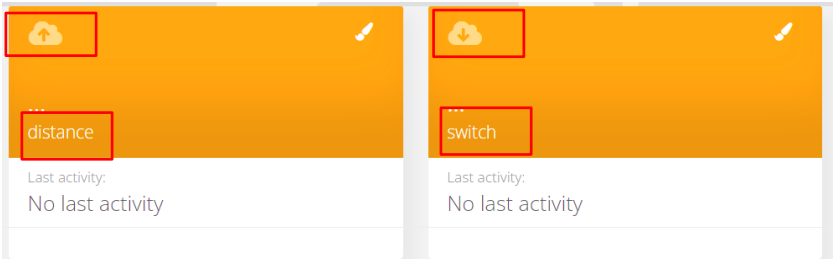
   https://ubidots.com/stem/

2. Login to your Ubidots account.
3. Create a new IoT device by selecting **Devices**>>  >>**Add New Device**>>**Blank Device** as shown in the picture below. Enter "**arduino**" for the device name.





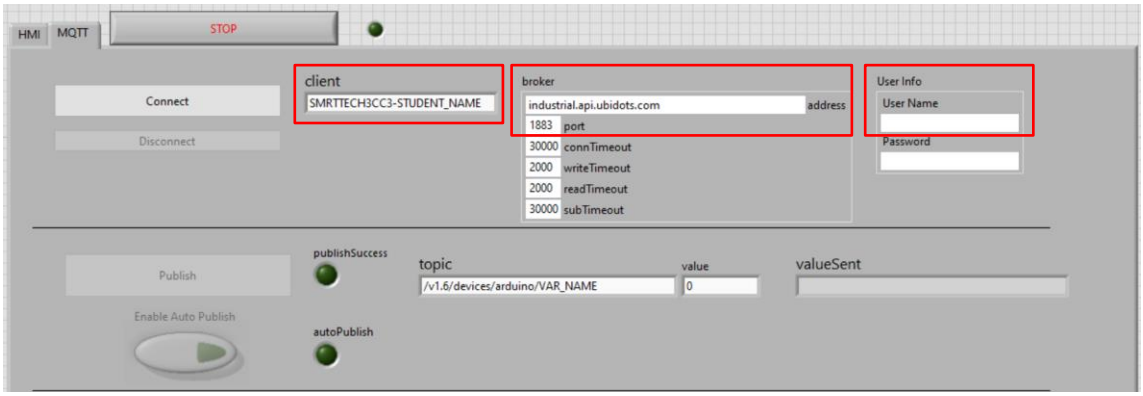
4. Click on the newly created IoT device and add two Raw variables (Raw) and name them **distance** and **switch** as shown below. You can change the icon of these variables by clicking on the variable icon.



5. In the next section we will use LabVIEW and MQTT to publish data into the distance variable and subscribe/read data from the switch variable.

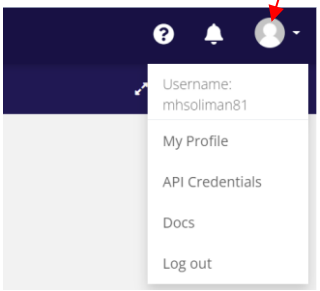
Exercise 2: Exchanging Data between LabVIEW and Ubidots MQTT broker

1. From avenue to learn download Lab7.zip files. Extract the zip file. Open the extracted folder and open LabVIEW file named “Lab7_init.vi”.
2. On the front panel, click the MQTT Tab and do the following

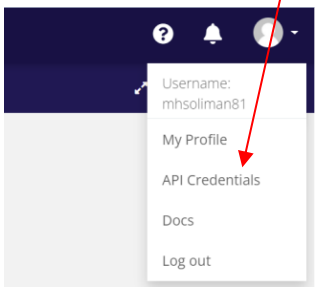


   a. Set the client’s name to a **unique name** for you. Replace “STUDENT_NAME” with your first and last name separated by “_”. Example: Mostafa_Soliman.
   b. Confirm the broker’s name to be **industrial.api.ubidots.com** and the port to **1883**
   c. Enter your **Ubidots token** into the **User Name**. You can copy your token from Ubidots as shown in the pictures below.

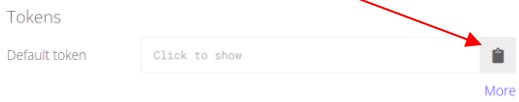
Click on your account icon



Click on your API Credentials



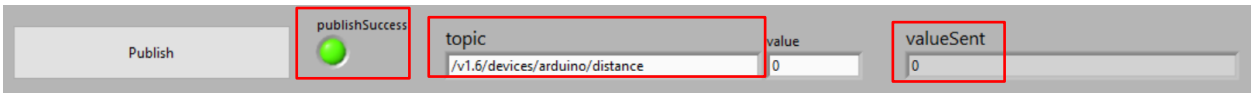
click copy token



3. In the publish section of the front panel, change the topic name to **/v1.6/devices/arduino/distance**

   Note: Ubidots MQTT broker requires the name of a published topic to have this form: /v1.6/devices/{DEVICE_LABEL}/{VAR_LABEL}

4. Run the LabVIEW program. Press connect, then press the “Publish” Push button and confirm there is a value sent.



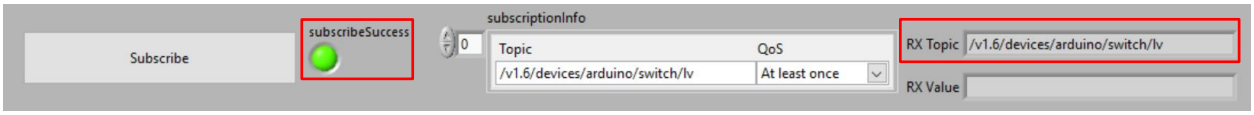
5. Go to Ubidots and confirm that the value is received by opening the variable detailed page (Devices>>Arduino>>distance)



6. Try couple of values, press publish, and confirm that the values are received by Ubidots. [Hint: you might need to refresh your page].
7. In the subscribe section of the front panel, change the topic name to **/v1.6/devices/arduino/switch/lv**

   Note: Ubidots MQTT broker requires the name of a subscribed topic to have this form: /v1.6/devices/{DEVICE_LABEL}/{ VARIABLE_LABEL}/lv

8. In the front panel, press the Subscribe button and that the subscribeSuccess LED turns green. You should also see the topic echoed in the RX Topic Indicator
9. Press the "Stop" Push button to close the communication ports and end the program.



IMPORTANT NOTES

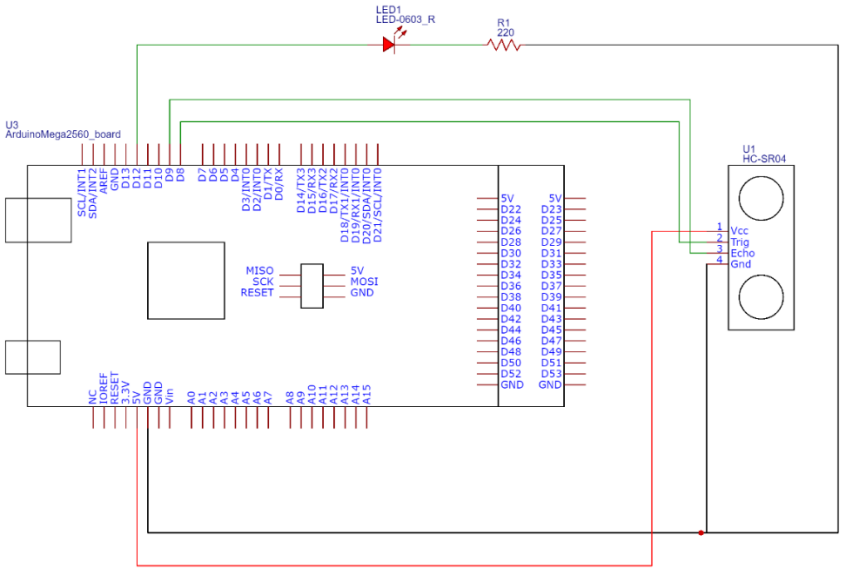
1. **Please do not stop your program by pressing the abort button  . You should always stop your program by pressing the "Stop" Push button.**
2. **IN THIS LAB, THE STOP BUTTON 

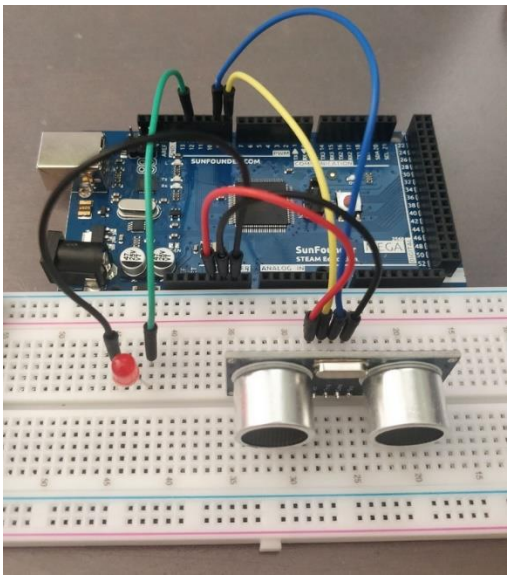
IN MAINTAINED. EVERYTIME, YOU WILL STOP THE PROGRAM BY PRESSING THE BUTTON, YOU MUST PRESS IT AGAIN BEFORE YOU RUN THE PROGRAM AGAIN.**

Exercise 3: Create a Circuit with an Ultrasonic Sensor and an LED

1. Connect **Digital (12)** of your Arduino board to an LED in series with 220 Ohm Resistor.
2. Place the **HC-SR04 Ultrasonic Sensor** into the **Breadboard**.
    a. Connect the **VCC** pin to the **Positive (+)** row.
    b. Connect the **GND** pin to the **Ground (-)** row.
    c. Connect the **Echo** pin to **Digital (9)**.
    d. Connect the **Trig** pin to **Digital (8)**.

Your circuit should now look like the figure below.



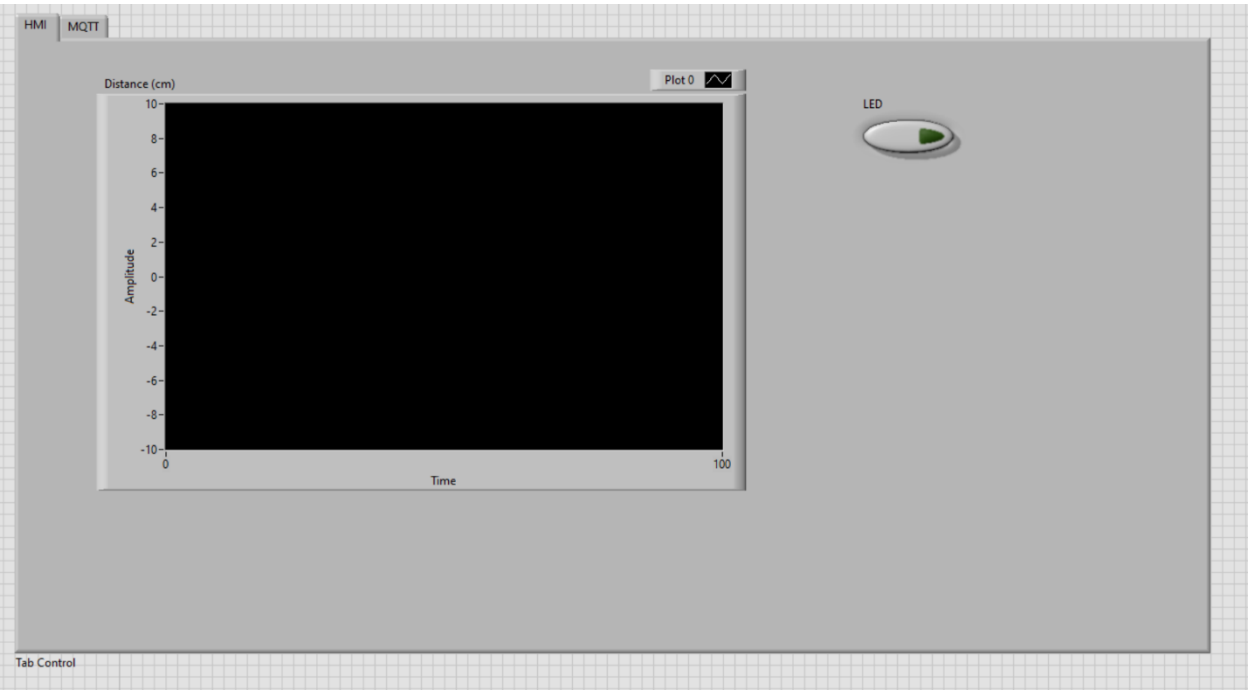


End of Exercise

Exercise 4: LabVIEW program creation

Front Panel

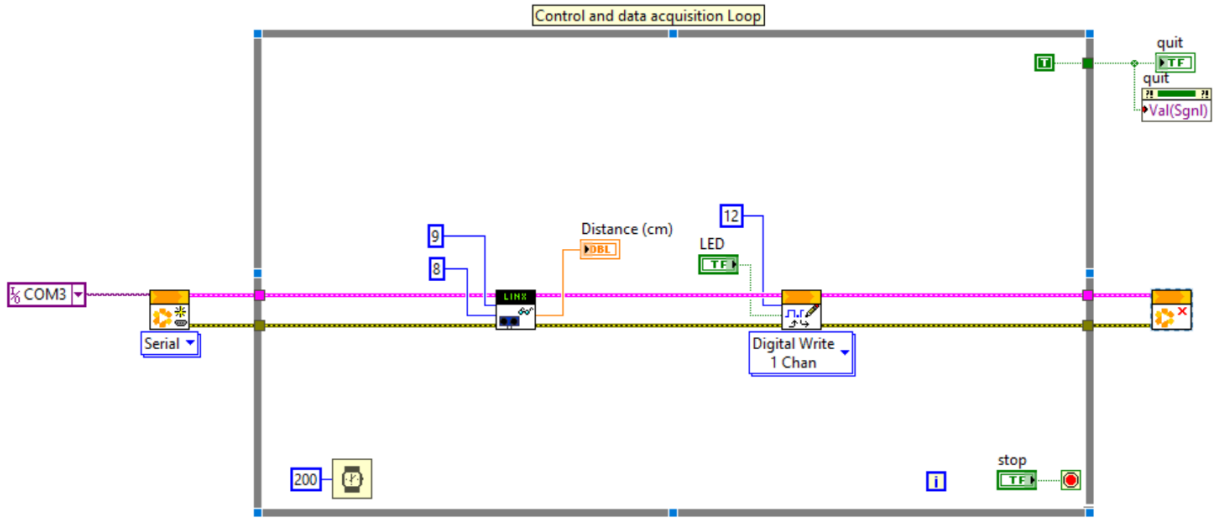
1. Open the front panel of your VI.
2. Add a **Waveform Chart** to the **HMI tab** from the **Controls»Graph** palette. Change the chart label to “Distance (cm)”.
3. Add a **Push Button** to the **HMI tab** from the **Controls»Boolean** palette. Change the Push Button label to “LED”.
4. You front panel should look like below.



Block Diagram

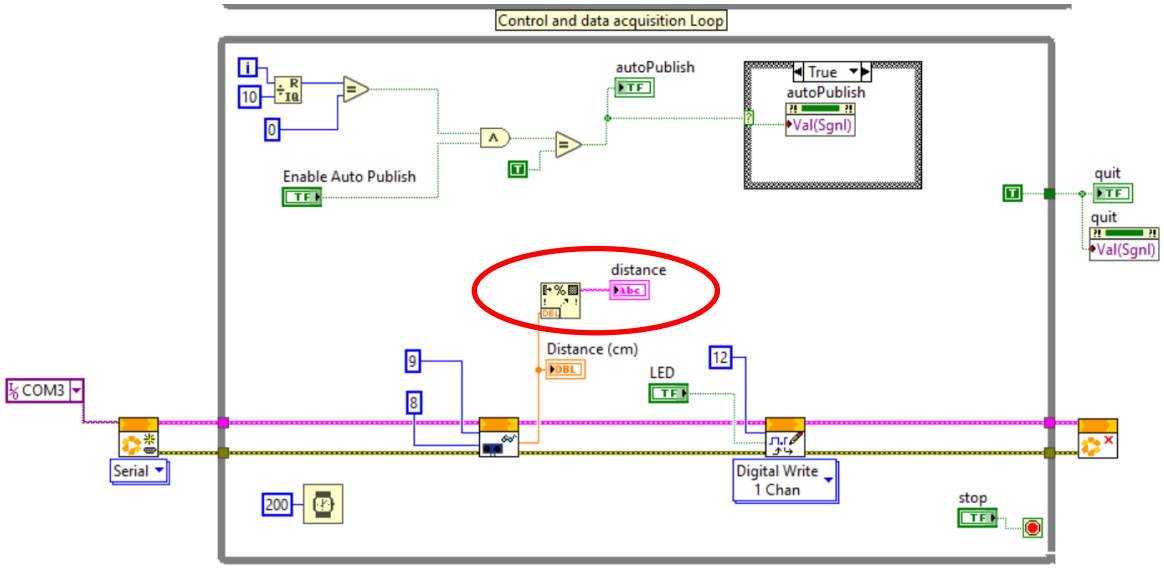
5. Open the block diagram of your VI.
6. Right click the Distance (cm) and the LED and uncheck **“View as Icon”** option.
7. Add **Linx Open**, **Linx Close**, **Digital Write** blocks from the **Linx Palette**.
8. Right click on the block diagram and select “**Select a VI..**” and select “**Ultrasonic Read Fixed**” file in the Lab7 Folder.
9. Wire all the blocks as shown below. Wire the `Centimetres` node to the **Distance (cm)** block and the **Digital Write** Output `Value` node to the **LED** block.
10. Add the required constants for each block by right clicking the block terminal and select create constant.
11. Run and test the LabVIEW Program.
    a. Confirm that you can toggle the LED from the front panel

b. Test the functioning of the ultrasonic sensor by putting a box or your hand in front of the sensor. Increase/decrease the distance between the box/hand and the sensor and observe the sensor readings on the chart.

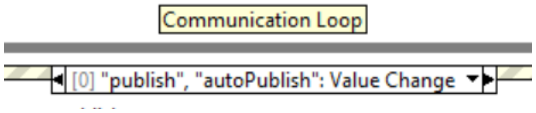


Exercise 5: Sending the Ultrasonic Sensor Data to the cloud

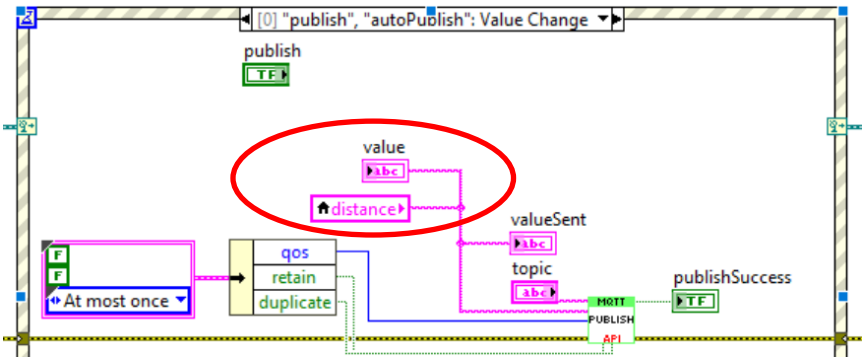
1. In the Control and data Acquisition Loop, add a **Format into String** function from the **Programming»Strings** palette. Right click the `Resulting String` node and select create Indicator. Rename the indicator block to `Distance` and uncheck the **"View as Icon"** option.



2. In the Event Structure of the top while loop, select the "publish","autoPublish" Case.



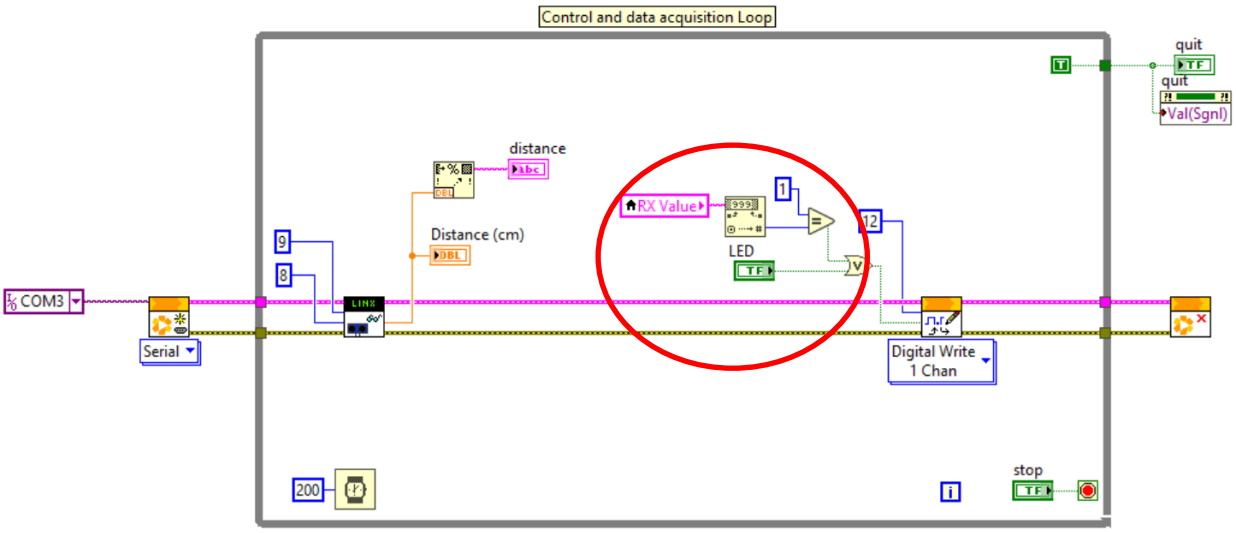
3. Replace the `Value` block with the `Distance` local variable.
   a. Right click the value block and select `change to indicator`.
   b. Add a **local variable** from **Programming»Structures palette**. Right click the variable and select "**Change to read**". Finally, click on the variable and select "Distance". The block diagram should look like the figure below.
   Why do we need to do this step?



4. Run LabVIEW. Click on the Connect and Enable Auto Publish push buttons. This will allow to send the sensor data every 1 second.
5. Go to Ubidots, open the distance variable detailed page (Devices>>Arduino>>distance), and confirm that the sensor values are continuously streamed to the variable. [Hint: you might need to refresh your page].

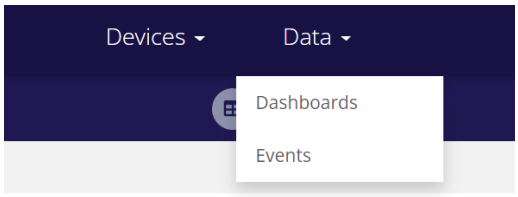
Exercise 6: Receiving Commands from the cloud

1. In the Control and data Acquisition Loop, right click on your block diagram. Press Search button. Enter "**Decimal String to Number**" and select the first block.
2. Add a **local variable** from **Programming»Structures palette**. Right click the variable and select "**Change to read**". Finally, click on the variable and select "**RX Value**".
3. Add **OR** and **Equal** functions and wire the blocks as shown below.

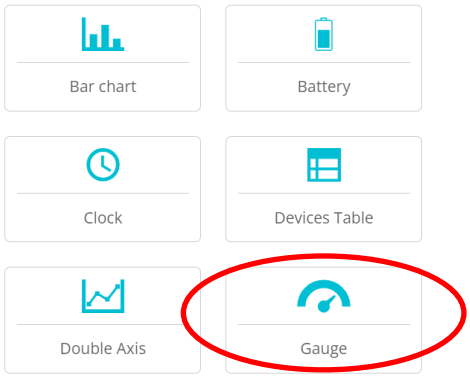


Exercise 7: Creating a web-based HMI (Dashboard)

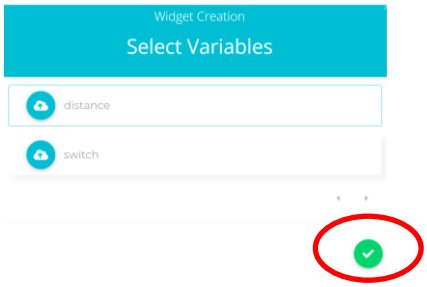
1. In Ubidots, click on data>>Dashboard.



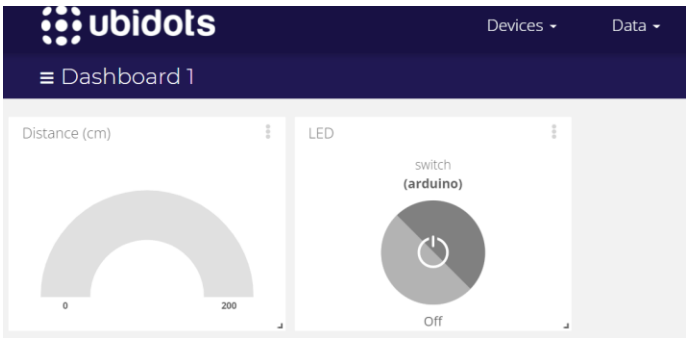
2. Click Add new Widget  and select a **Gauge**.



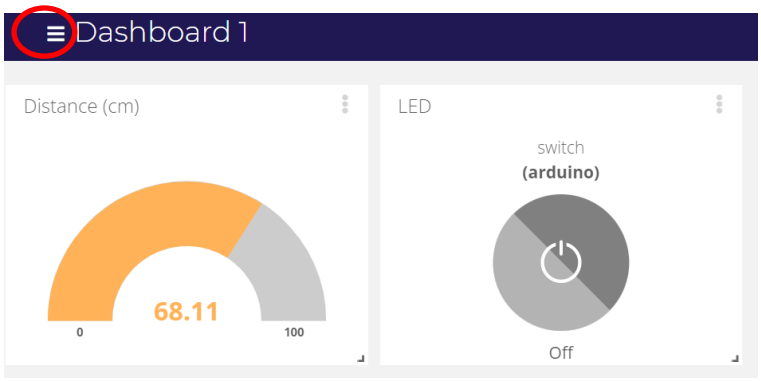
3. Select **Add Variables**>>**arduino**>>**distance** and press on the green check mark.



4. In the appearance section, set the name to “Distance (cm)”.
5. Like the previous steps, add a Switch widget and add the variable arduino>>switch to it. Set the name of the switch to “LED”
6. Your dashboard should look like this



7. Run your LabVIEW program, press the publish, subscribe, and Enable Auto Publish buttons.
8. Confirm that the ultrasonic measurements are read on the gauge.
9. Confirm that you can turn the LED on and off using the switch.
10. Click on the dashboard menu button. Click on the share button . Copy your dashboard URL.



11. Demonstrate your dashboard to your lab instructor by providing him/her with the URL and confirm remote monitoring and control functionalities.
12. Optional: You might open the dashboard from your phone using your dashboard URL and confirm that the dashboard will scale properly.

EXTEND YOUR KNOWLEDGE

1. Modify your code to automatically turn the LED when the distance is less than 5 cm.

SUBMISSION

1. Draw a block diagram of the system that you built and configured in this lab. In your diagram, use arrows to show how the data are transferred between the web-based HMI and the sensors and actuators.
2. Use your own words to describe the operation of the ultrasonic sensor. What is the purpose of the Trig and Echo pins.
3. For the application that you built, mention any advantage of using LabVIEW for programing the application instead of Arduino IDE.
4. Include the new code corresponding to the extend your knowledge section.
5. Include any typo that you found or suggestion for improvement for this lab manual.